

Aarhus University, Department of Mathematics
Bachelor's Degree Programme in Mathematics-Economics
May Re-exam 2022/2023
Written Re-exam in: 550171U026 Investment and Finance
Duration: 4 hours
Open book examination, all supplementary material allowed, no internet

Exercise 1 (weight 25%)

Assume that the term structure of interest rates is flat in all scenarios. The table reports the prices of a bond and an option at three different level of interest rates.

Rate (%)	Bond	Option
2.50	99.7573	3.3755
2.70	99.0190	2.9659
2.90	98.4904	2.6471

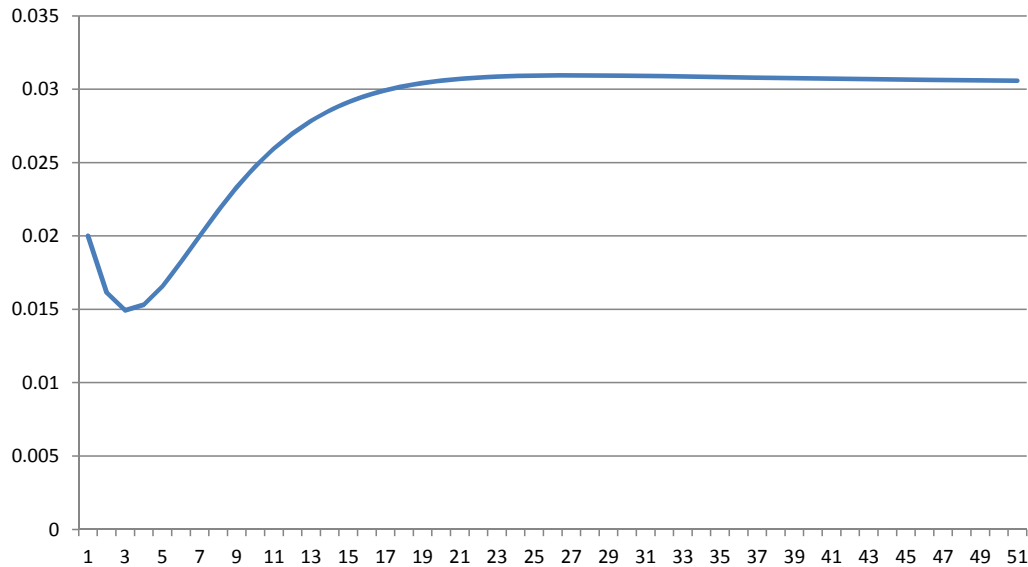
1. Give a numerical estimate of the sensitivity to an interest rate change of the two assets at a rate of 2.70%.
2. Assume that you have bought the option and that the rate is 2.70%. Can you perform a local hedge of the option using the bond? (Explain in detail)
3. Assume that the rate decreases by 20 basis points. What is the profit-loss of the hedged position?
4. Give an estimate of the modified duration of the two assets when the rate is at 2.70%.

Answer the following questions independently of the first four questions.

5. Compute the price of a perpetuity with semi-annual payments of 10 and yield-to-maturity equal to 5% under semi-annual compounding.
6. Compute the modified duration of the perpetuity introduced in the previous question.
7. On a given day you observe the depicted spot rate term structure. You want to fit the Nelson-Siegel parametrization to the observed term structure.

$$s_t = \phi_0 + \phi_1 e^{-\frac{t}{\phi_4}} + \frac{\phi_2 t}{\phi_4} e^{-\frac{t}{\phi_4}} + \phi_3 \left(\frac{\left(1 - e^{-\frac{t}{\phi_4}}\right)}{\frac{t}{\phi_4}} - \left(1 + \frac{1}{2} \frac{t}{\phi_4}\right) e^{-\frac{t}{\phi_4}} \right)$$

Which values would you suggest for the parameters ϕ_0 and ϕ_1 if you had to make a qualified guess without doing any calibrations?



Exercise 2 (weight 25%)

Consider the stocks X and Y with covariances with the market portfolio $\text{Cov}(r_X, r_M) = 0.01$ and $\text{Cov}(r_Y, r_M) = 0.02$, respectively. The market portfolio has expected return $\bar{r}_M = 0.1625$ and variance $\text{Var}(r_M) = 0.04$. The riskfree rate equals $r_f = 0.095$. Answer the following questions:

1. Compute the CAPM beta and the expected returns of the two stocks according to the CAPM.
2. Consider a portfolio with weights in the two assets equal to $w_X = 0.25$ and $w_Y = 0.75$ and with the price 89.50. You assume that the portfolio can be sold for the price of 100 after a year. Should you invest in the portfolio?
3. What would be your answer if instead the portfolio weights were $w_X = 0.75$ and $w_Y = 0.25$?
4. Assume that the 2-factor APT model holds and that the two factors have risk premiums $\lambda_1 = 0.06$ and $\lambda_2 = 0.08$. Moreover, assume that the portfolio's sensitivity to the two factors are $b_1 = 0.25$ and $b_2 = 0.125$. What is the expected return according to the APT model?
5. Give an interpretation of the risk premiums in the APT model.
6. Determine the portfolio weights that ensure that the CAPM and the APT model are aligned.

Exercise 3

Consider a state-space model with two periods: Today ($t = 0$) and tomorrow ($t = 1$). The financial market consists of two assets d^1 and d^2 and three possible states $s \in \{1, 2, 3\}$. The possible values of the assets tomorrow are given by the vectors

$$\begin{pmatrix} d_1^1 & d_2^1 & d_3^1 \end{pmatrix} = \begin{pmatrix} 10 & 15 & 22 \end{pmatrix}$$
$$\begin{pmatrix} d_1^2 & d_2^2 & d_3^2 \end{pmatrix} = \begin{pmatrix} 8 & 17 & 9 \end{pmatrix}$$

The prices of the assets today are, respectively, $P_1 = 10.5$ and $P_2 = 8.9$.

1. Argue that the vector $\psi = \begin{pmatrix} 0.3 & 0.25 & 0.25 \end{pmatrix}$ is a vector of state-prices for this model and that the model is free of arbitrage. Is the market also complete?
2. What is the risk-free rate implied by the state prices ψ ?
3. Assume that a call option on asset 1 with strike $K = 12$ is traded in the market. What price would you quote for this call option?
4. A third asset d^3 is introduced to the market, with possible payments

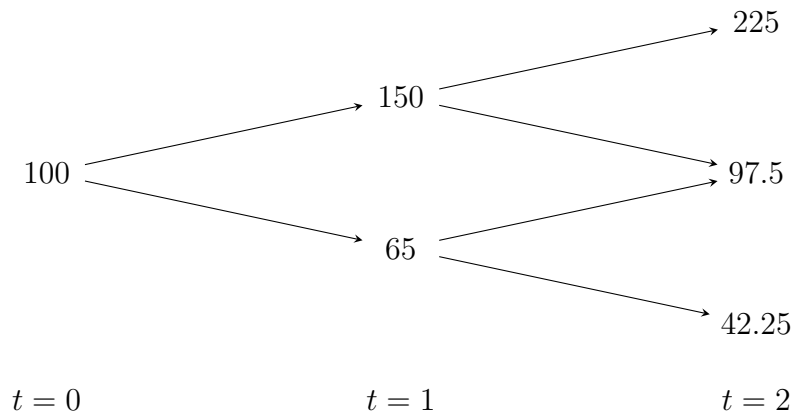
$$d^3 = \begin{pmatrix} 5 & 15 & 20 \end{pmatrix}$$

what is the price of this asset, P_3 , given state prices ψ ?

5. It turns out that the price of asset 3 is observed in the market as $P_3 = 7$. Is this consistent with no arbitrage?

Exercise 4

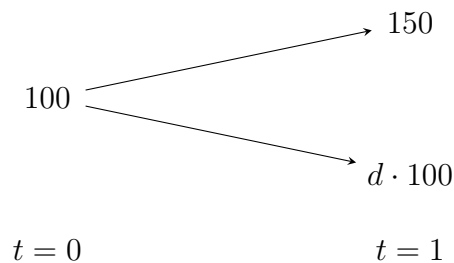
Consider the following binomial lattice model for the price evolution of a non-dividend paying stock S over two periods:



The constant, discretely compounded risk-free rate is $r = 5\%$.

1. What is the risk-neutral probability of ending up in state $S(2) = 97.5$?
2. What is the price of an American call and put option on S with strike $K = 85$ and maturity $T = 2$?

Assume now that the first period, the price evolves as



where $d \geq 0$.

3. Determine the interval $[\underline{d}, \bar{d}]$ with $\underline{d} \leq d \leq \bar{d}$ such that the model is free of arbitrage.
4. Determine the possible interval of European call prices $[\underline{C}, \bar{C}]$ with strike $K = 85$ given that $d \in [\underline{d}, \bar{d}]$.

Assume now that the stock price follows a geometric Brownian motion:

$$dS(t) = \mu S(t)dt + \sigma S(t)dW_t$$

where W_t is a Brownian motion and $\mu, \sigma > 0$. The interest rate r is still constant, but now *continuously* compounded.

5. A European call option with strike $K = 100$ and maturity $T = 1.25$ years is trading with an implied volatility of 0.35. What is the price of this option?
6. There is also a derivative on the market with payoff function $\sqrt{S(T)}$. You are told that the pricing function F of this derivative is given as

$$F_t = \sqrt{S(t)} e^{-\frac{1}{2}(r+\sigma^2)(T-t)}$$

is this pricing function consistent with no arbitrage?